

# YIBO LIN

Associate Professor ◇ Department of Design Automation and Computing System

School of Integrated Circuits, 5 Yiheyuan Road, Beijing, China, 100871 ◇ Peking University

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## RESEARCH INTERESTS

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Modeling and optimization in VLSI CAD, machine learning applications, and heterogeneous computing

## EDUCATION

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**University of Texas at Austin, Texas, USA** Aug. 2013 – May 2018

Ph.D., Department of Electrical and Computer Engineering

**Shanghai Jiao Tong University, Shanghai, China** Sep. 2009 – Jun. 2013

B.S., Department of Microelectronics

## EXPERIENCE

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**Peking University, Beijing, China** Aug. 2025 – present

Associate Professor

**Peking University, Beijing, China** Jul. 2019 – Jul. 2025

Assistant Professor

Department of Design Automation and Computing System,

School of Integrated Circuits (since Nov. 2021)

Center for Energy-efficient Computing and Applications (CECA),

School of Electronics Engineering and Computer Science

**University of Texas at Austin, Texas, U.S.** Jun. 2018 – Jun. 2019

Postdoc

Department of Electrical and Computer Engineering

## TEACHING EXPERIENCE

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Practice in AI-Assisted VLSI Circuit Design

Instructor Graduate 2026

Fundamental Algorithms for Engineering in Integrated Circuits

Instructor Graduate 2023–2026

Design Automation and Computing System

Instructor Graduate 2022–2024

Optimization and Machine Learning in VLSI Design Automation

Instructor Undergraduate 2021–2024

Introduction to Computing B

Instructor Undergraduate 2020–2022

## AWARDS AND HONORS

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|---------------------------------------------------------|-------------------------------|------|
| Best Paper Award Nomination (19/266)                    | ICCAD                         | 2025 |
| Best Paper Award Nomination (7/420)                     | DAC                           | 2025 |
| Honorable Mention (×2) (/135)                           | ISEDA                         | 2025 |
| Best Paper Award Nomination (13/168)                    | ASPDAC                        | 2025 |
| Best Paper Award Nomination (10/195)                    | ICCAD                         | 2024 |
| Best Paper Award (6/151) & Honorable Mention            | ISEDA                         | 2024 |
| Early Career Award (only one per year)                  | CCF Technical Committee in IC | 2023 |
| Inaugural Best Reviewer Award                           | ICCAD                         | 2023 |
| Best Paper Award (4/205)                                | DATE                          | 2023 |
| Best Paper Award (4/249)                                | DATE                          | 2022 |
| Best Paper Award Nomination (6/132)                     | ICCAD                         | 2022 |
| Donald O. Pederson Best Paper Award (2/3495 in 4 years) | TCAD                          | 2021 |
| Best Paper Award                                        | ISPD                          | 2020 |
| Best Paper Award Nomination (12/86)                     | ASPDAC                        | 2020 |
| Best Paper Award (1/201) & Nomination (5/201)           | DAC                           | 2019 |
| Best Paper Award Nomination                             | ISPD                          | 2019 |
| Inaugural Best Paper Award                              | Integration, the VLSI Journal | 2018 |
| Franco Cerrina Memorial Best Student Paper Award        | SPIE                          | 2016 |
| A. Richard Newton Young Student Fellow                  | DAC                           | 2014 |

## PROFESSIONAL SERVICE

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### Executive Committee Member

- General co-chair, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD) 2025–2026
- Program co-chair, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD) 2024
- Panel chair, IEEE International Symposium of Electronics Design Automation (ISEDA) 2023–2024
- Financial chair, ACM/IEEE Workshop on Machine Learning for CAD (MLCAD) 2021–2023

### Journal Editor

- Associate Editor, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024 – present
- Associate Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES), 2024 – present
- Associate Editor, Elsevier Integration, the VLSI Journal (Integration), 2024 – present
- Guest Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2024
- Guest Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2022

### Contest Chair

- Contest Problem Chair, Interated Circuits EDA Elite Challenge 2021 – 2023

## Technical Program Committee Member

- ACM/IEEE Design Automation Conference (DAC): 2020
- IEEE/ACM International Conference on Computer-Aided Design (ICCAD): 2018 – 2021, 2023
- Design, Automation and Test in Europe Conference (DATE): 2025
- IEEE International Conference on Computer Design (ICCD): 2019
- IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC): 2021 – 2022
- ACM International Symposium on Physical Design (ISPD): 2020, 2025
- ACM/IEEE Workshop on Machine Learning for CAD (MLCAD): 2021
- ACM Great Lakes Symposium on VLSI (GLVLSI): 2024
- Workshop on Synthesis And System Integration of Mixed Information technologies (SASIMI): 2021
- IEEE Electron Devices Technology and Manufacturing Conference (EDTM): 2021
- IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS): 2022.

## Journal Reviewer

- IEEE Transaction on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Computers (TC)
- ACM Transaction on Design Automation of Electronic Systems (TODAES)
- SPIE Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3)
- Elsevier, Integration, the VLSI Journal (Integration)

## Volunteer

- ACM SIGDA Website Administrator, 2021 – present

## 10 SELECTED PUBLICATIONS

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Inverse chronological order. \* denotes corresponding authors.

1. Yifan Chen, Zaiwen Wen, Yun Liang and **Yibo Lin\***, “[Stronger Mixed-Size Placement Backbone Considering Second-Order Information](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct 29-31, 2023.
2. Haoyi Zhang, Xiaohan Gao, Haoyang Luo, Jiahao Song, Xiyuan Tang, Junhua Liu, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023. (**Best Paper Award**)
3. Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin\***, “[Multi-Electrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity, Clock Feasibility, and Timing Optimization](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2023.
4. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin\***, “[Accelerating Static Timing Analysis using CPU-GPU Heterogeneous Parallelism](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2023.

5. Qipan Wang, Xiaohan Gao, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[DeePEB: A Neural Partial Differential Equation Solver for Post Exposure Baking Simulation in Lithography](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022. (**Best Paper Nomination**)
6. Zizheng Guo and **Yibo Lin\***, “[Differentiable-Timing-Driven Global Placement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.
7. Siting Liu, Peiyu Liao, Zhitang Chen, Wenlong Lv, **Yibo Lin\*** and Bei Yu\*, “[FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022. (**Best Paper Award**)
8. Zizheng Guo, Jing Mai and **Yibo Lin\***, “[Ultrafast CPU/GPU Kernels for Density Accumulation in Placement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
9. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin\***, “[A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
10. **Yibo Lin\***, Zixuan Jiang, Jiaqi Gu, Wuxi Li, Shounak Dhar, Haoxing Ren, Brucek Khailany and David Z. Pan, “[DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2021. (**Best Paper Award**)

## PUBLICATIONS

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### Book Chapters

- [B3] **Yibo Lin**, Zizheng Guo and Jing Mai, “[Deep Learning Framework for Placement](#)”, Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)
- [B2] Haoyu Yang, **Yibo Lin** and Bei Yu, “[Machine Learning for Mask Synthesis and Verification](#)”, Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)

### Conference and Journal Papers (\* denotes corresponding authors)

*Summary: DAC (37), ICCAD (24), IEEE TCAD (33), DATE (22), etc.*

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- [J238] Zhenkun Lin, **Yibo Lin**, Genggeng Liu\* and Gang Du, “[URoute: Universal Routability Prediction](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)
- [J237] Xu He\*, Yushan Wang, Chenjing Yang, Renjun Zhao, **Yibo Lin**, Peiyu Liao, Bei Yu, Yao Wang, Chang Liu and Yang Guo, “[Efficient Timing Prediction and Optimization Using Derivable Gradient Boosting Machine Model at Placement Stage](#)”, ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026. (accepted)
- [J236] Chunyuan Zhao, Zizheng Guo, Zuodong Zhang and **Yibo Lin\***, “[GAP-LA: GPU-Accelerated Performance-Driven Layer Assignment](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

- [J235] Xufeng Yao, Jiayi Jiang, Yuxuan Zhao, Peiyu Liao, **Yibo Lin** and Bei Yu\*, “EvoPlace: Evolution of Optimization Algorithms for Global Placement via Large Language Models”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)
- [C234] Zixuan Pan, Zizheng Guo, Yufan Du, Runsheng Wang and **Yibo Lin**\*, “Disentangled Differentiable Timing-Power Co-Optimization with Quad-Gradient Gate sizing”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (accepted)
- [C233] Yi Ren, Yukun Wang, Xiang Meng, Guoyao Cheng, Baokang Peng, Lining Zhang, **Yibo Lin**, Runsheng Wang and Guangyu Sun\*, “CellE: Automated Standard Cell Library Extension via Equality Saturation”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (accepted)
- [C232] Jing Mai, Yi-Chen Lu, **Yibo Lin** and Haoxing Ren, “LATTE: Legality-Assured Differentiable Timing-Driven Detailed Placement”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (accepted)
- [C231] Wenbo An, Kairong Guo, Haoyi Zhang and **Yibo Lin**\*, “EPiCell: Electro-Physical Co-Modeling for Standard Cell PPA Prediction”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (accepted)
- [C230] Yuhao Ji, Yunqi Shi, Tianshu Hou, Yuxuan Zhao, Chunyuan Zhao, Peiyu Liao, Zizheng Guo, Chao Qian, **Yibo Lin** and Bei Yu\*, “Integrated Timing-driven Placement for Hybrid-Bonding-based Face-to-Face 3D ICs”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (accepted)
- [C229] Kexing Zhou, Youwei Zhuo, **Yibo Lin**, Weikang Qian, Pengpeng Ren and Yun Liang\*, “Activity-Aware Partitioning for Effective Multi-Threaded Event-Driven RTL Simulation”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (accepted)
- [C228] Jiechen Huang, Hector R. Rodriguez, Dingcheng Yang, Zuochang Ye, **Yibo Lin** and Wenjian Yu\*, “AttentionCap: Transformer Based Capacitance Matrix Learning Toward Full-Chip Extraction”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (accepted)
- [C227] Haoyi Zhang, Weijian Fan, Xiaohan Gao, Bingyang Liu, Runsheng Wang and **Yibo Lin**\*, “PANDA: An LLM-Enhanced Performance-Driven Analog Design Framework Bridging Design Intent and Layout Generation”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (accepted)(**Invited Paper**)
- [C226] Xizhe Shi, Zizheng Guo, Zuodong Zhang, Yun Liang, Runsheng Wang\* and **Yibo Lin**\*, “HeteroPower: A CPU-GPU Heterogeneous Engine to Accelerate Gate-Level Power Analysis”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)
- [C225] Kairong Guo, Jiechen Huang, Wenjian Yu and **Yibo Lin**\*, “CAPCell: Standard Cell Layout Synthesis with Parasitic Capacitance Aware Parallel Sampling”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.
- [C224] Jincheng Lou, Ruohan Xu, Jiecheng Ma, Runzhe Tao, Xinyu Qu and **Yibo Lin**\*, “LEGO: An LLM Skill-Based Front-End Design Generation Platform”, IEEE/ACM International Sym-

- posium of EDA (ISED), Singapore, May 8-12, 2026. (**Best Paper Nomination**)
- [C223] Zizhuo Fu, Yifan Zhou, Zhaoxin Lu, Guangyu Sun, Runsheng Wang, Meng Li\* and **Yibo Lin\***, “RePart: Efficient Hypergraph Partitioning with Logic Replication Optimization for Multi-FPGA System”, IEEE/ACM International Symposium of EDA (ISED), Singapore, May 8-12, 2026.
- [C222] Xun Jiang, Haoran Lu, Yifan Chen, Jiarui Wang, Chunyuan Zhao, Yibo Zhang, Jianxiang Jin, Heng Wu, Runsheng Wang and **Yibo Lin\***, “Flexible Double-Side Pin Redistribution with Efficient Virtual-Net-Guided Cell Type Assignment”, IEEE/ACM International Symposium of EDA (ISED), Singapore, May 8-12, 2026. (**Best Paper Nomination**)
- [C221] Haichuan Liu, Zizheng Guo, Runsheng Wang and **Yibo Lin\***, “ElmoreCeff: A GPU-Friendly Elmore-Like Delay Calculator with a Closed-Form Effective Capacitance Model”, IEEE/ACM International Symposium of EDA (ISED), Singapore, May 8-12, 2026.
- [C220] Bingyang Liu, Haoning Jiang, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, David Z. Pan and **Yibo Lin\***, “GRAIN: A Design-Intent-Driven Analog Layout Migration Framework”, IEEE/ACM International Symposium of EDA (ISED), Singapore, May 8-12, 2026.
- [C219] Mingjun Wang, Yihan Wen, Yuntao Lu, Fengrui Liu, Yuxiang Zhao, Boyu Han, Jianan Mu, **Yibo Lin**, Runsheng Wang, Huawei Li and Bei Yu, “[CircuitNet 3.0: A Multi-Modal Dataset with Task-Oriented Augmentation for AI-Driven Circuit Design](#)”, International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, Apr 23-27, 2026. (accepted)
- [C218] Mengjia Dai, Hongduo Liu, Yuhao Ji, Xiaojing Su, **Yibo Lin** and Bei Yu\*, “Soft-Constrained Triple Patterning Layout Decomposition”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C217] Chenglin Ye, Yuze Lu, Yizhan Liu, Ligong Zhang, Jinghan Xu, Fei Liu, **Yibo Lin**, Zheng Zhou and Xiaoyan Liu, “EMAPER: Cross-Level Electromigration Aware Placement and Routing EDA Workflow for Interconnects Hotspot Prediction and Mitigation”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C216] Jiajun Tan, Qichao Ma, Yiming Du, Yiming Gan, Ling Liang, **Yibo Lin**, Ming Zhu, Zongwei Wang and Yimao Cai, “SONIC: Smart Optimization for Neural-Integrated CMP with Timing-Aware Fills”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C215] Zizheng Guo, Runsheng Wang and **Yibo Lin\***, “Provably Optimal Planar Pareto Nearest Neighbor Search with Double Monotone Chains”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C214] Jianqing Chen, Zhenkun Lin, Xun Jiang, Genggeng Liu\*, **Yibo Lin** and Gang Du, “DSR: A Systematic Approach for Efficient Double-Sided Signal Routing”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C213] Siting Liu, Peng Xu, Peiyu Liao, Keren Zhu, **Yibo Lin** and Bei Yu\*, “Submodular Maximization-Inspired Adaptive Routing Bend Space Planning”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C212] Tianxiang Zhu, Qipan Wang, **Yibo Lin\*** and Runsheng Wang\*, “Efficient Warpage Simulation Of Complex 2.5-D/3-D IC Structures with Novel Meshing Algorithm and Layerwise

- Plate Theory”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C211] Jiarui Wang, Runzhe Tao, Jing Mai, Xun Jiang, Shenghua Wang, Cuiliu Yang, Haoyu Jie, Kan Huang, Richard Y. Sun and **Yibo Lin\***, “TDM Signal Grouping and Package Pin Assignment for 2.5D Multi-FPGA Systems with Lookahead Placement”, International Symposium on Field-Programmable Gate Arrays (FPGA), Monterey, CA, Feb 22-24, 2026.
- [C210] Zizheng Guo, Haichuan Liu, Xizhe Shi, Shenglu Hua, Zuodong Zhang, Chunyuan Zhao, Runsheng Wang and **Yibo Lin\***, “HeteroSTA: A CPU-GPU Heterogeneous Static Timing Analysis Engine with Holistic Industrial Design Support”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026. (**Invited Paper**)
- [C209] Xizhe Shi, Zizheng Guo, **Yibo Lin\***, Zuodong Zhang, Yun Liang and Runsheng Wang\*, “HeteroLatch: A CPU-GPU Heterogeneous Latch-Aware Timing Analysis Engine”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [C208] Kairong Guo, Haoran Lu, Rui Guo, Jiarui Wang, Chunyuan Zhao, Heng Wu, Runsheng Wang and **Yibo Lin\***, “Standard Cell Layout Synthesis for Dual-Sided 3D-Stacked Transistors”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [C207] Weijian Fan, Haoyi Zhang, Weibin Lin, Runsheng Wang and **Yibo Lin\***, “MOSTAR: Multi-Stage Hierarchical Bayesian Optimization for Substructure-Aware High-Dimensional Analog Circuit Sizing”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [C206] Chenhao Xue, Kezhi Li, Jiaxing Zhang, Yi Ren, Zhengyuan Shi, Chen Zhang, **Yibo Lin**, Lining Zhang, Qiang Xu and Guangyu Sun\*, “AC-Refiner: Efficient Arithmetic Circuit Optimization Using Conditional Diffusion Models”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [C205] Jingchao Hu, **Yibo Lin**, Hao Yu, Quan Chen, Zhou Jin\* and Cheng Zhuo, “IR Drop-Aware ECO: A Fast Approach to Minimize Layout and Timing Disturbance”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [J204] Siting Liu, Ziyi Wang, Fangzhou Liu, **Yibo Lin**, Bei Yu\* and Martin Wong, “[Sign-off Timing Considerations via Concurrent Routing Topology Optimization](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [J203] Zhenkun Lin, Genggeng Liu\*, Xing Huang, **Yibo Lin**, Jixin Zhang, Wenhao Liu and Ting-Chi Wang, “[A Unified Deep Reinforcement Learning Approach for Constructing Rectilinear and Octilinear Steiner Minimum Tree](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [J202] Jingchen Zhu, Chenhao Xue, Yiqi Chen, Zhao Wang, Chen Zhang, Yu Shen, Yifan Chen, Zekang Cheng, Yu Jiang, Tianqi Wang, **Yibo Lin**, Wei Hu, Bin Cui, Runsheng Wang, Yun Liang and Guangyu Sun\*, “[Theseus: Exploring Efficient Wafer-Scale Chip Design for Large Language Models](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [J201] Yajuan Su, Zixi Liu, **Yibo Lin**, Xiaojing Su, Yuqin Wang, Xin Hong, Yujie Jiang, Pengyu Ren and Yayi Wei, “[A Post-Routing Layout Optimization Framework for Lithography Process](#)

- [Window Enlargement](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [J200] Jing Mai, Chunyuan Zhao, Zuodong Zhang, Zhixiong Di, Runsheng Wang and **Yibo Lin\***, “[LEGALM 2.0: A Versatile Augmented Lagrangian Method-Based Methodology for Mixed-Cell-Height Legalization](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [C199] Chunyuan Zhao, Jiarui Wang, Xun Jiang, Jincheng Lou and **Yibo Lin\***, “GTA: GPU-Accelerated Track Assignment with Lightweight Lookup Table for Conflict Detection”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025. (**Best Paper Nomination**)
- [C198] Yufan Du, Zizheng Guo, Runsheng Wang and **Yibo Lin\***, “Differentiable Physical Optimization”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C197] Yuhao Ji, Yuntao Lu, Zuodong Zhang, Zizheng Guo, **Yibo Lin** and Bei Yu\*, “DiffCCD: Differentiable Concurrent Clock and Data Optimization”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C196] Yuan Pu, Yuhao Ji, Siying Yu, Zuodong Zhang, Zizheng Guo, Zhuolun He, **Yibo Lin**, David Z. Pan and Bei Yu\*, “GPU Acceleration for Versatile Buffer Insertion”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C195] Yi Ren, Baokang Peng, Chenhao Xue, Kairong Guo, Yukun Wang, Guoyao Cheng, **Yibo Lin**, Lining Zhang and Guangyu Sun\*, “Orthrux: Dual-Loop Automated Framework for System-Technology Co-Optimization”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C194] Haichuan Liu, Zizheng Guo, Runsheng Wang and **Yibo Lin\***, “IncreGPUSTA: GPU-Accelerated Incremental Static Timing Analysis for Iterative Design Flows”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C193] Tianxiang Zhu, Qipan Wang, **Yibo Lin\*** and Runsheng Wang\*, “High-Resolution Full-Chip Thermal Resistance Extraction of BEOL Interconnects in 3-D ICs Considering Detailed Via Connectivity”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C192] Chenchen Zhao, Zhengyuan Shi, Xiangyu Wen, Chengjie Liu, Yi Liu, Yunhao Zhou, Yuxiang Zhao, Hefei Feng, Yinan Zhu, Waa Gwok-Wan, Xin Cheng, Weiyu Chen, Yongqi Fu, Chujie Chen, Chenhao Xue, Ying Wang, **Yibo Lin**, Jun Yang, Ning Xu, Xi Wang and Qiang Xu\*, “MMCircuitEval: A Comprehensive Multimodal Circuit-Focused Benchmark for Evaluating LLMs”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [J191] Bingyang Liu, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[LayoutCopilot: An LLM-powered Multi-agent Collaborative Framework for Interactive Analog Layout Design](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2025.
- [C190] Yuxiang Zhao, Zhuomin Chai, Xun Jiang, Qiang Xu, Runsheng Wang and **Yibo Lin\***, “[DeepLayout: Learning Neural Representations of Circuit Placement Layout](#)”, International



Conference on Machine Learning (ICML), Vancouver, Canada, Jul 13-19, 2025.

- [C189] Lijie Zeng, Jiatai Sun, Xiao Wu, Dan Niu, Tianshi Wang, **Yibo Lin**, Zuochang Ye and Zhou Jin\*, “[G-SpNN: GPU-Accelerated Passivity Enforcement for S-Parameter Modeling with Neural Networks](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [C188] Ziyang Yu, Peng Xu, Zixiao Wang, Binwu Zhu, Qipan Wang, **Yibo Lin**, Runsheng Wang, Bei Yu\* and Martin Wong, “[SDM-PEB: Spatial-Depthwise Mamba for Enhanced Post-Exposure Bake Simulation](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [C187] Zizheng Guo, Yanqing Zhang, Runsheng Wang, **Yibo Lin** and Haoxing Ren, “[GEM: GPU-Accelerated Emulator-Inspired RTL Simulation](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025. (**Best Paper Nomination**)
- [C186] Xun Jiang, Haoran Lu, Yuxuan Zhao, Jiarui Wang, Zizheng Guo, Heng Wu, Bei Yu, Sung Kyu Lim, Runsheng Wang, Ru Huang and **Yibo Lin**\*, “[A Systematic Approach for Multi-Objective Double-Side Clock Tree Synthesis](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [C185] Jiarui Wang, Yanjing Liu and **Yibo Lin**\*, “[Synergistic Die-Level Router for Multi-FPGA System with Time-Division Multiplexing Optimization](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [C184] Yifan Chen, Jing Mai, Zuodong Zhang and **Yibo Lin**\*, “[RUPlace: Optimizing Routability via Unified Placement and Routing Formulation](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [J183] Yuxiang Zhao, Zhuomin Chai, Xun Jiang, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[PDNNet: PDN-Aware GNN-CNN Heterogeneous Network for Dynamic IR Drop Prediction](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2025.
- [J182] Rui Guo, Haoran Lu, Jiacheng Sun, Xun Jiang, Lining Zhangm, Ming Li, **Yibo Lin**, Runsheng Wang, Heng Wu\* and Ru Huang, “[Design Optimization of Flip FET Standard Cells With Dual-Sided Pins for Ultimate Scaling](#)”, IEEE Transactions on Electron Devices (TED), Jun, 2025.
- [C181] Haoyi Zhang, Shizhao Sun, **Yibo Lin**, Runsheng Wang and Jiang Bian, “[AnalogXpert: Automating Analog Topology Synthesis by Incorporating Circuit Design Expertise into Large Language Models](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025.
- [C180] Yufan Du, Zizheng Guo, Yang Hsu, Zhili Xiong, Seunggeun Kim, David Z. Pan, Runsheng Wang and **Yibo Lin**\*, “[Addressing Continuity and Expressivity Limitations in Differentiable Physical Optimization: A Case Study in Gate Sizing](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025. (**Honorable Mention Paper Award**)
- [C179] Kairong Guo and **Yibo Lin**\*, “[Multi-Row Standard Cell Layout Synthesis with Enhanced Scalability](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025.

- [C178] Qipan Wang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[ATSim3.5D: A Multiscale Thermal Simulator for 3.5D-IC Systems based on Nonlinear Multigrid Method](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025. (**Honorable Mention Paper Award**)
- [C177] Chenhao Xue, Yi Ren, Jinwei Zhou, Kezhi Li, Chen Zhang, **Yibo Lin**, Lining Zhang, Qiang Xu and Guangyu Sun\*, “[DOMAC: Differentiable Optimization for High-Speed Multipliers and Multiply-Accumulators](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025.
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- [C59] Wei Ye, Mohamed Baker Alawieh, Yuki Watanabe, Shigeki Nojima, **Yibo Lin** and David Z. Pan\*, “[TEMPO: Fast Mask Topography Effect Modeling with Deep Learning](#)”, ACM International Symposium on Physical Design (ISPD), Taipei, Taiwan, Sep 20-23, 2020. (**Best Paper Award**)
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- [C51] Rachel Selina Rajarathnam, **Yibo Lin**, Yier Jin and David Z. Pan\*, “[ReGDS: A Reverse Engineering Framework from GDSII to Gate-level Netlist](#)”, IEEE International Workshop on Hardware-Oriented Security and Trust (HOST), San Jose, CA, May 4, 2020.
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- [C49] Mohamed Baker Alawieh, Wuxi Li, **Yibo Lin**, Love Singhal, Mahesh Iyer and David Z. Pan\*, “[High-Definition Routing Congestion Prediction for Large-Scale FPGAs](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jan 13-16, 2020.
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- [J44] **Yibo Lin**, Meng Li, Yuki Watanabe, Taiki Kimura, Tetsuaki Matsunawa, Shigeki Nojima and David Z. Pan\*, “[Data Efficient Lithography Modeling with Transfer Learning and Active Data Selection](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2019.
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- [C41] Wei Ye, Mohamed Baker Alawieh, **Yibo Lin** and David Z. Pan\*, “[LithoGAN: End-to-End Lithography Modeling with Generative Adversarial Networks](#)”, ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019. (**Best Paper Nomination**)
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